

✓ DAY 8 Q2

```
from tensorflow.keras.datasets import cifar10

(x_train, y_train), (x_test, y_test) = cifar10.load_data()
```

```
x_valid = x_train[700:900]
y_valid = y_train[700:900]
x_train = x_train[:700]
y_train = y_train[:700]
x_test = x_test[:200]
y_test = y_test[:200]
```

```
def normalize(data):
    data = data.astype("float32") / 255.0
    return data

x_train = normalize(x_train)
x_test = normalize(x_test)
x_valid = normalize(x_valid)
y_train = normalize(y_train)
y_test = normalize(y_test)
y_valid = normalize(y_valid)
```

```
print(x_train.shape, y_train.shape)
print(x_test.shape, y_test.shape)
print(x_valid.shape, y_valid.shape)
```

```
(700, 32, 32, 3) (700, 1)
(200, 32, 32, 3) (200, 1)
(200, 32, 32, 3) (200, 1)
```

```
print(f"Training Sample {x_train.shape[0]}")
print(f"Testing Sample {x_test.shape[0]}")
print(f"Validation Sample {x_valid.shape[0]}")
```

```
Training Sample 700  
Testing Sample 200  
Validation Sample 200
```